Table 5-6: Examples of Fourier transform pairs. Note that constant $a \geq 0$.

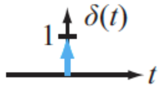
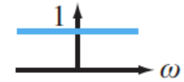
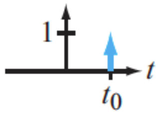
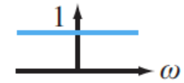
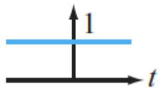
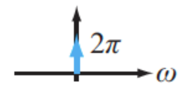
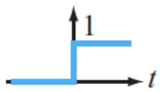
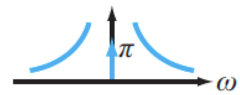
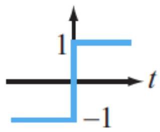
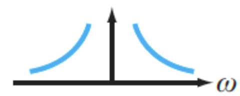
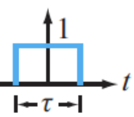
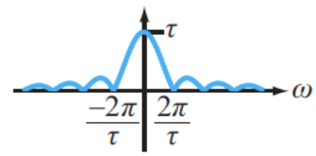
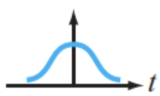
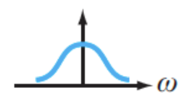
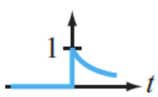
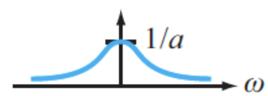
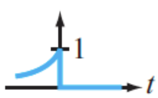
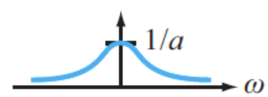
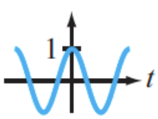
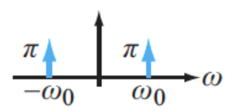
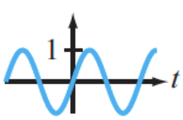
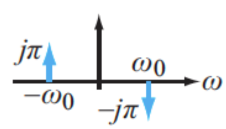
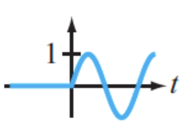
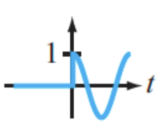
	$x(t)$	$\mathbf{X}(\omega) = \mathcal{F}[x(t)]$	$\lvert\mathbf{X}(\omega)\rvert$
		BASIC FUNCTIONS	
1.		$\delta(t) \longleftrightarrow 1$	
1a.		$\delta(t-t_0) \longleftrightarrow e^{-j\omega t_0}$	
2.		$1 \longleftrightarrow 2\pi\ \delta(\omega)$	
3.		$u(t) \longleftrightarrow \pi\ \delta(\omega) + 1/j\omega$	
4.		$\mathrm{sgn}(t) \longleftrightarrow 2/j\omega$	
5.		$\mathrm{rect}(t/\tau) \longleftrightarrow \tau\ \mathrm{sinc}(\omega\tau/2)$	
6.		$\dfrac{e^{-t^2/(2\sigma^2)}}{\sqrt{2\pi\sigma^2}} \longleftrightarrow e^{-\omega^2\sigma^2/2}$	
7a.		$e^{-at}\ u(t) \longleftrightarrow 1/(a+j\omega)$	
7b.		$e^{at}\ u(-t) \longleftrightarrow 1/(a-j\omega)$	
8.		$\cos\omega_0 t \longleftrightarrow \pi[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]$	
9.		$\sin\omega_0 t \longleftrightarrow j\pi[\delta(\omega+\omega_0)-\delta(\omega-\omega_0)]$	
		ADDITIONAL FUNCTIONS	
10.		$e^{j\omega_0 t} \longleftrightarrow 2\pi\ \delta(\omega-\omega_0)$	
11.		$te^{-at}\ u(t) \longleftrightarrow 1/(a+j\omega)^2$	
12a.		$[e^{-at}\sin\omega_0 t]\ u(t) \longleftrightarrow \omega_0/[(a+j\omega)^2+\omega_0^2]$	
12b.		$[\sin\omega_0 t]\ u(t) \longleftrightarrow (\pi/2j)[\delta(\omega-\omega_0)-\delta(\omega+\omega_0)]+[\omega_0^2/(\omega_0^2-\omega^2)]$	
13a.		$[e^{-at}\cos\omega_0 t]\ u(t) \longleftrightarrow (a+j\omega)/[(a+j\omega)^2+\omega_0^2]$	
13b.		$[\cos\omega_0 t]\ u(t) \longleftrightarrow (\pi/2)[\delta(\omega-\omega_0)+\delta(\omega+\omega_0)]+[j\omega/(\omega_0^2-\omega^2)]$	

Table 5-7: Major properties of the Fourier transform.

Property	$x(t)$		$\mathbf{X}(\omega)=\mathcal{F}[x(t)]=\int_{-\infty}^{\infty} x(t)\, e^{-j\omega t}\, dt$
1. Multiplication by a constant	$K\, x(t)$	$\leftrightarrow$	$K\, \mathbf{X}(\omega)$
2. Linearity	$K_1\, x_1(t)+K_2\, x_2(t)$	$\leftrightarrow$	$K_1\, \mathbf{X}_1(\omega)+K_2\, \mathbf{X}_2(\omega)$
3. Time scaling	$x(at)$	$\leftrightarrow$	$\frac{1}{\lvert a\rvert}\, \mathbf{X}\left(\frac{\omega}{a}\right)$
4. Time shift	$x(t-t_0)$	$\leftrightarrow$	$e^{-j\omega t_0}\, \mathbf{X}(\omega)$
5. Frequency shift	$e^{j\omega_0 t}\, x(t)$	$\leftrightarrow$	$\mathbf{X}(\omega-\omega_0)$
6. Time 1st derivative	$x'=\frac{dx}{dt}$	$\leftrightarrow$	$j\omega\, \mathbf{X}(\omega)$
7. Time *n*th derivative	$\frac{d^n x}{dt^n}$	$\leftrightarrow$	$(j\omega)^n\, \mathbf{X}(\omega)$
8. Time integral	$\int_{-\infty}^{t} x(\tau)\, d\tau$	$\leftrightarrow$	$\frac{\mathbf{X}(\omega)}{j\omega}+\pi\, \mathbf{X}(0)\, \delta(\omega)$, where $\mathbf{X}(0)=\int_{-\infty}^{\infty} x(t)\, dt$
9. Frequency derivative	$t^n\, x(t)$	$\leftrightarrow$	$(j)^n \frac{d^n \mathbf{X}(\omega)}{d\omega^n}$
10. Modulation	$x(t)\, \cos\omega_0 t$	$\leftrightarrow$	$\frac{1}{2}[\mathbf{X}(\omega-\omega_0)+\mathbf{X}(\omega+\omega_0)]$
11. Convolution in t	$x_1(t) * x_2(t)$	$\leftrightarrow$	$\mathbf{X}_1(\omega)\, \mathbf{X}_2(\omega)$
12. Convolution in ω	$x_1(t)\, x_2(t)$	$\leftrightarrow$	$\frac{1}{2\pi}\, \mathbf{X}_1(\omega) * \mathbf{X}_2(\omega)$
13. Conjugate symmetry			$\mathbf{X}(-\omega)=\mathbf{X}^*(\omega)$